RAMS JLCPCB bare ESP32-S3 wearable

Engineering review: confirm BOM, assembly and bring-up gates before fabrication release.

01 Power and USB

File: RAMS_JLC_S3_power.kicad_sch

02 ESP32-S3 and 2.4 GHz RF

File: RAMS_JLC_S3_mcu.kicad_sch

03 LoRa and I-O

File: RAMS_JLC_S3_io.kicad_sch

RAMS

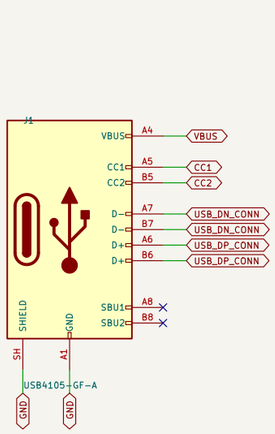
Sheet: /
File: RAMS_JLC_S3.kicad_sch

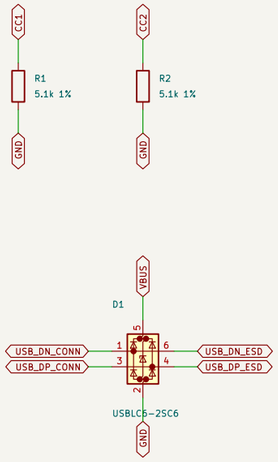
Title: RAMS JLCPCB bare ESP32-S3 wearable

Size: A3	Date: 2026-11-09	**Rev: J2 LDO**
KiCad E.D.A. 10.0.6		Id: 1/4

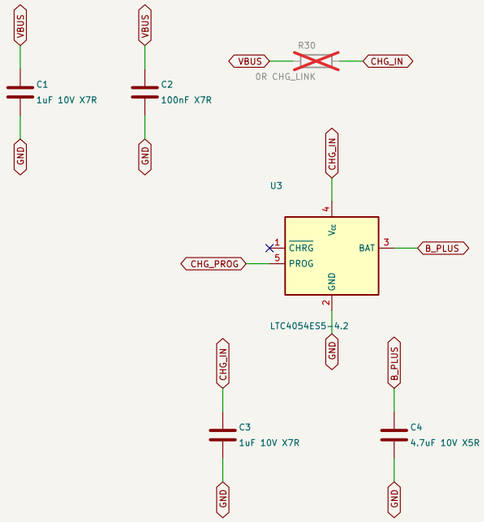
01 Power and USB

Single-cell LiPo. CHG_LINK DNP pending cell/harness check. AP2112 3.3 V LDO; no switching converter.

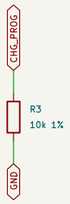




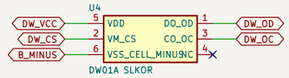
USB-C input and ESD

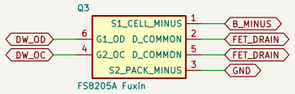


100 mA charger; R30 initially DNP







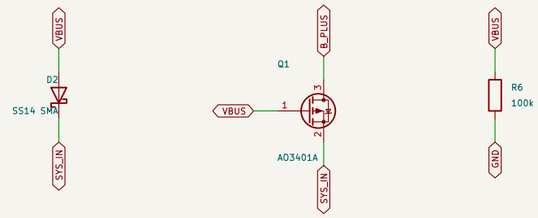




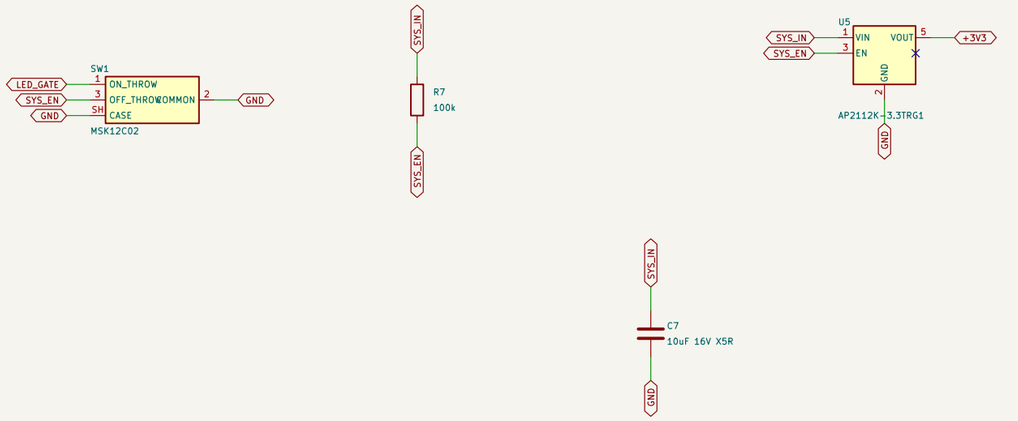


Cell B_MINUS is separate from protected system GND

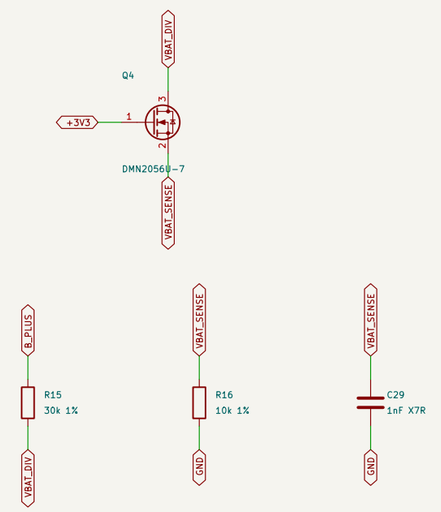
USB / battery load sharing



3.3 V linear regulator (owned AP2112)



4:1 battery ADC; Q4 gate follows 3V3



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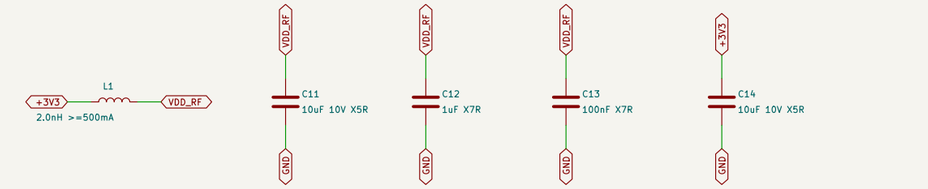
Sheet: /01 Power and USB/
File: RAMS_JLC_S3_power.kicad_sch

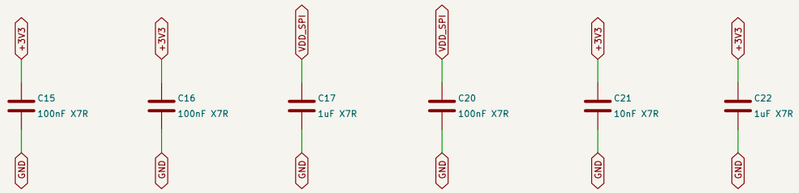
Title: 01 Power and USB

Size: A3	Date: 2026-11-09	**Rev: J2 LDO**
KiCad E.D.A. 10.0.6		Id: 2/4

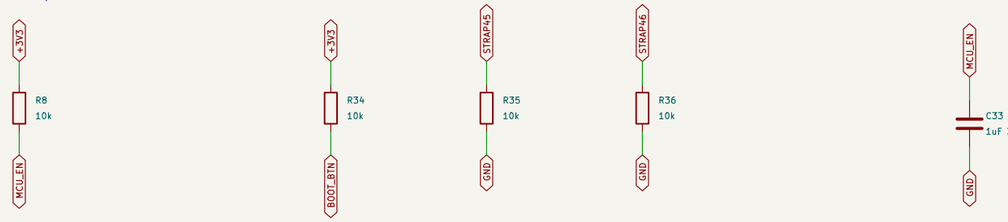
02 ESP32-S3 and 2.4 GHz RF

RF reference values require tuning with the assembled board and final enclosure.





Enable, boot and strap bias

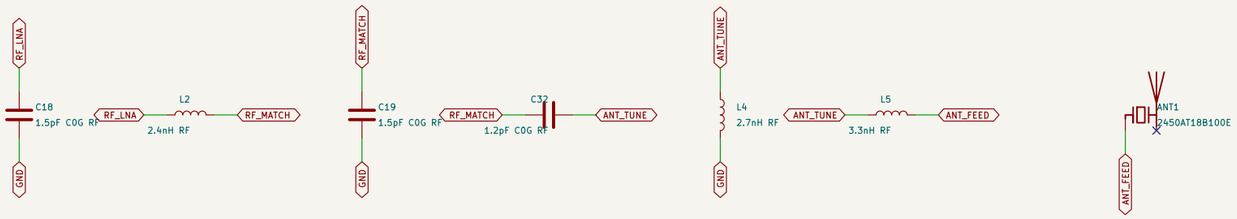


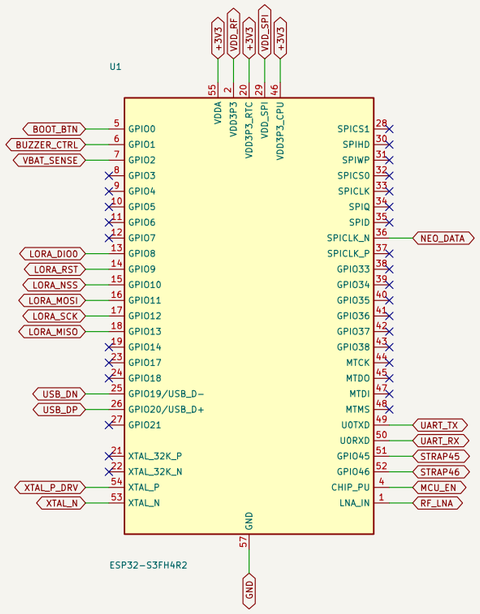
40 MHz crystal oscillator





Chip match, feed and antenna tuning





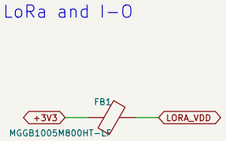
RAMS

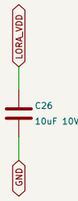
Sheet: /02 ESP32-S3 and 2.4 GHz RF/
File: RAMS_JLC_S3_mcu.kicad_sch

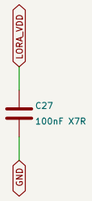
Title: 02 ESP32-S3 and 2.4 GHz RF

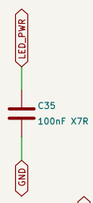
Size: A3	Date: 2026-11-09	**Rev: J2 LDO**
KiCad E.D.A. 10.0.6		Id: 3/4

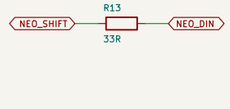
03 LoRa and I-O

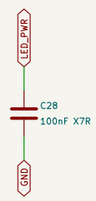




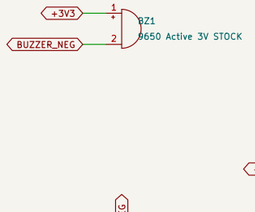




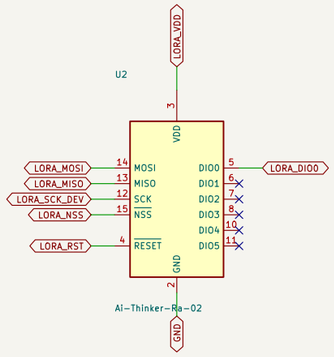




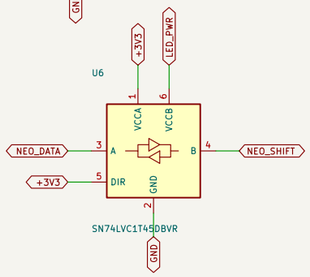
Phone supplies IMU/GPS over BLE. Hand-solder bottom Ra-02 after top-side reflow.

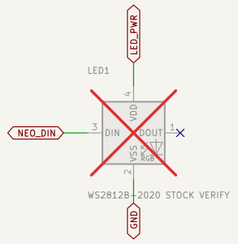


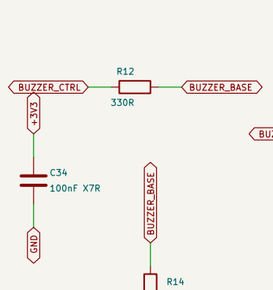


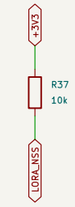


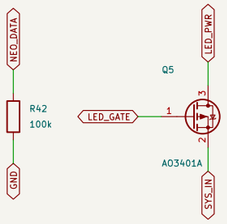




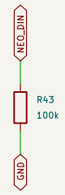












433 MHz LoRa radio

2020 RGB: switched SYS_IN supply + data translation

Verify exact LED pinout, timing and low-battery operation before fitting.

Active buzzer and low-side driver

Reset / boot-cancel buttons and USB series resistors









ERC source declarations: physical sources identified in circuit specification



















Bring-up test pads

















RAMS

Sheet: /03 LoRa and I-O/
File: RAMS_JLC_S3_io.kicad_sch

Title: 03 LoRa and I-O

Size: A3	Date: 2026-11-09	**Rev: J2 LDO**
KiCad E.D.A. 10.0.6		Id: 4/4